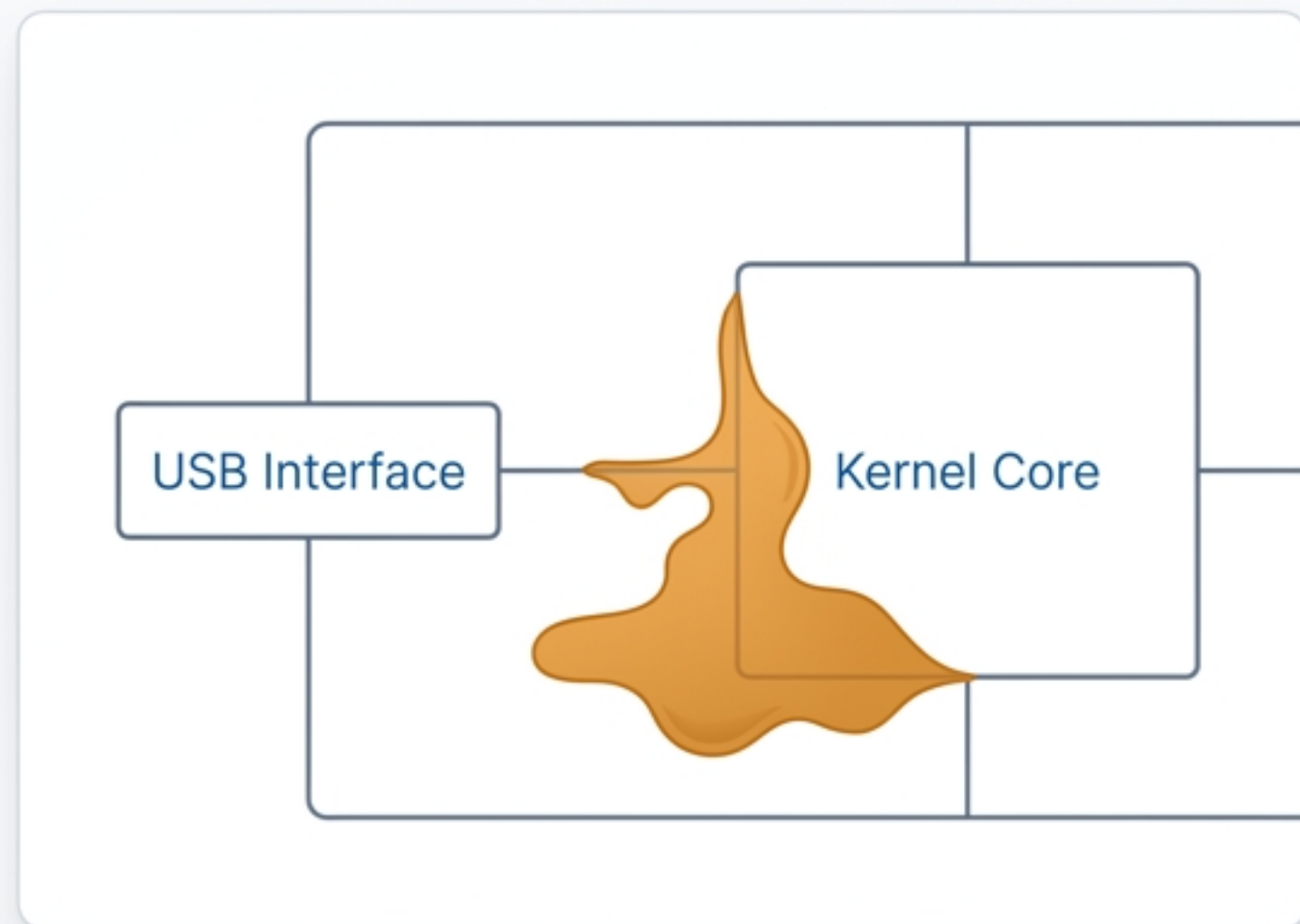


# CVE-2026-23037

Memory Leak in ETAS ES58X Linux Driver:  
Vulnerability Autopsy & Remediation



Identity

**CVE-2026-23037**

**CVE-2026-23037**

Class

CWE-401 (Missing  
Release of Memory)

Severity

 Low-Medium

Component

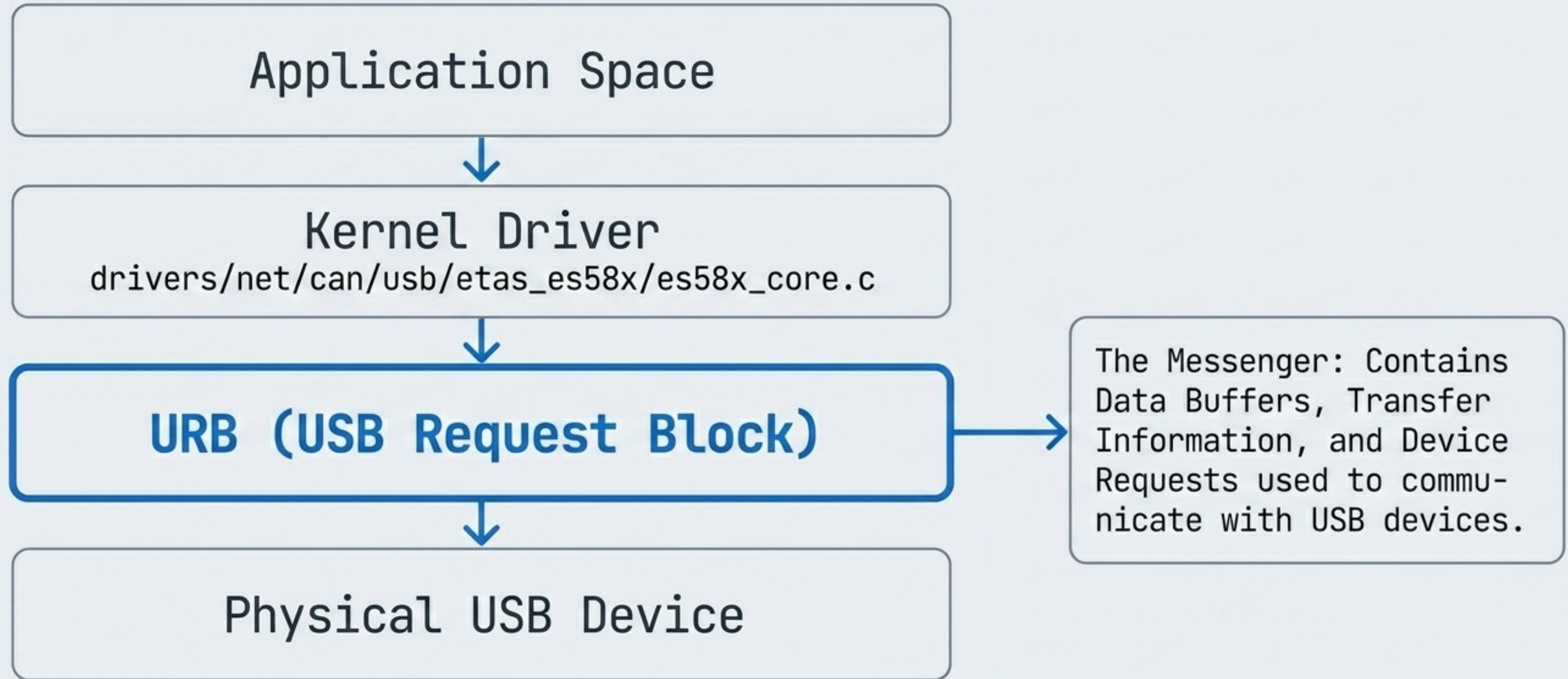
**Linux Kernel (ETAS ES58X  
CAN USB Driver)**

Primary Risk

**Resource  
Exhaustion**



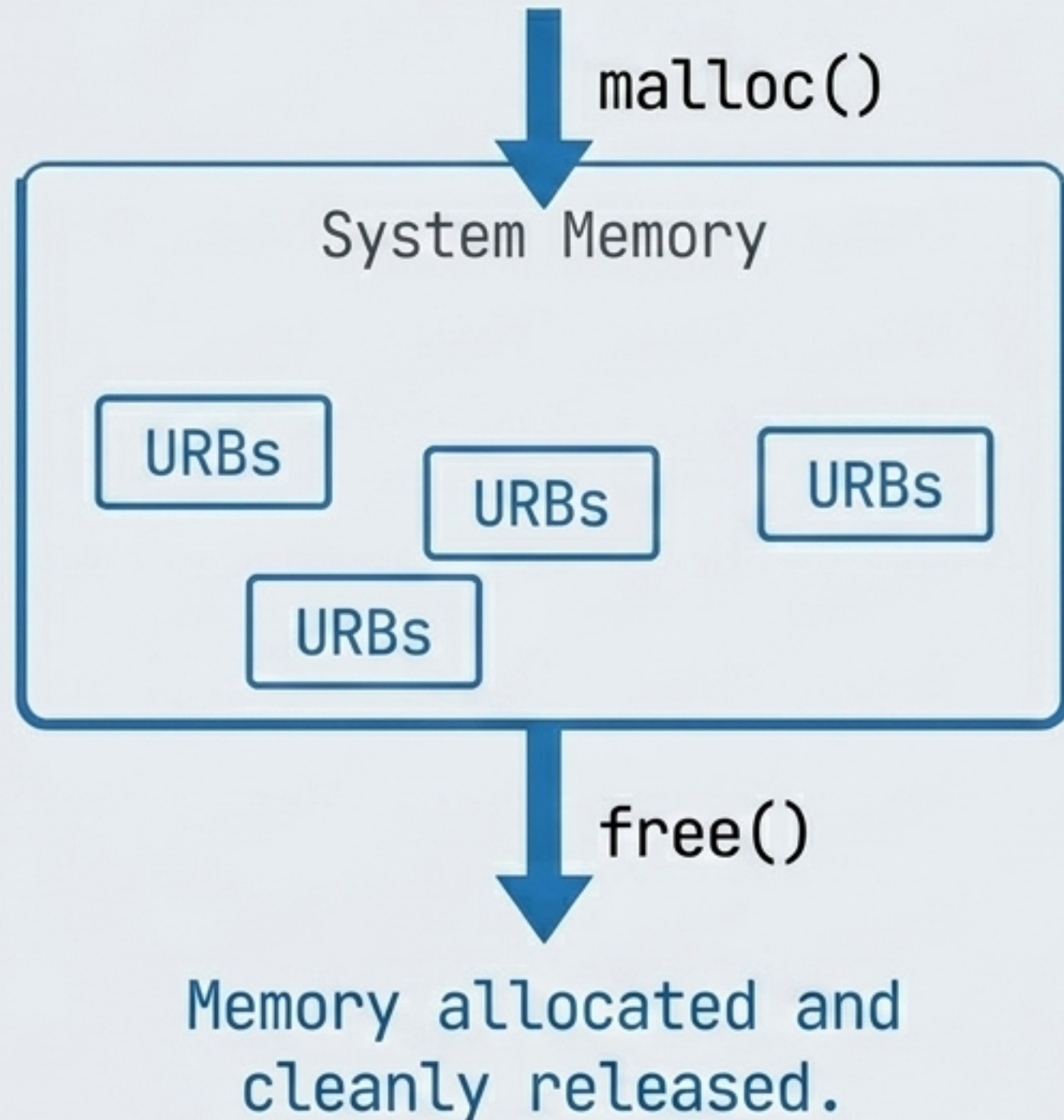
# Where the Flaw Lives: The URB Stack



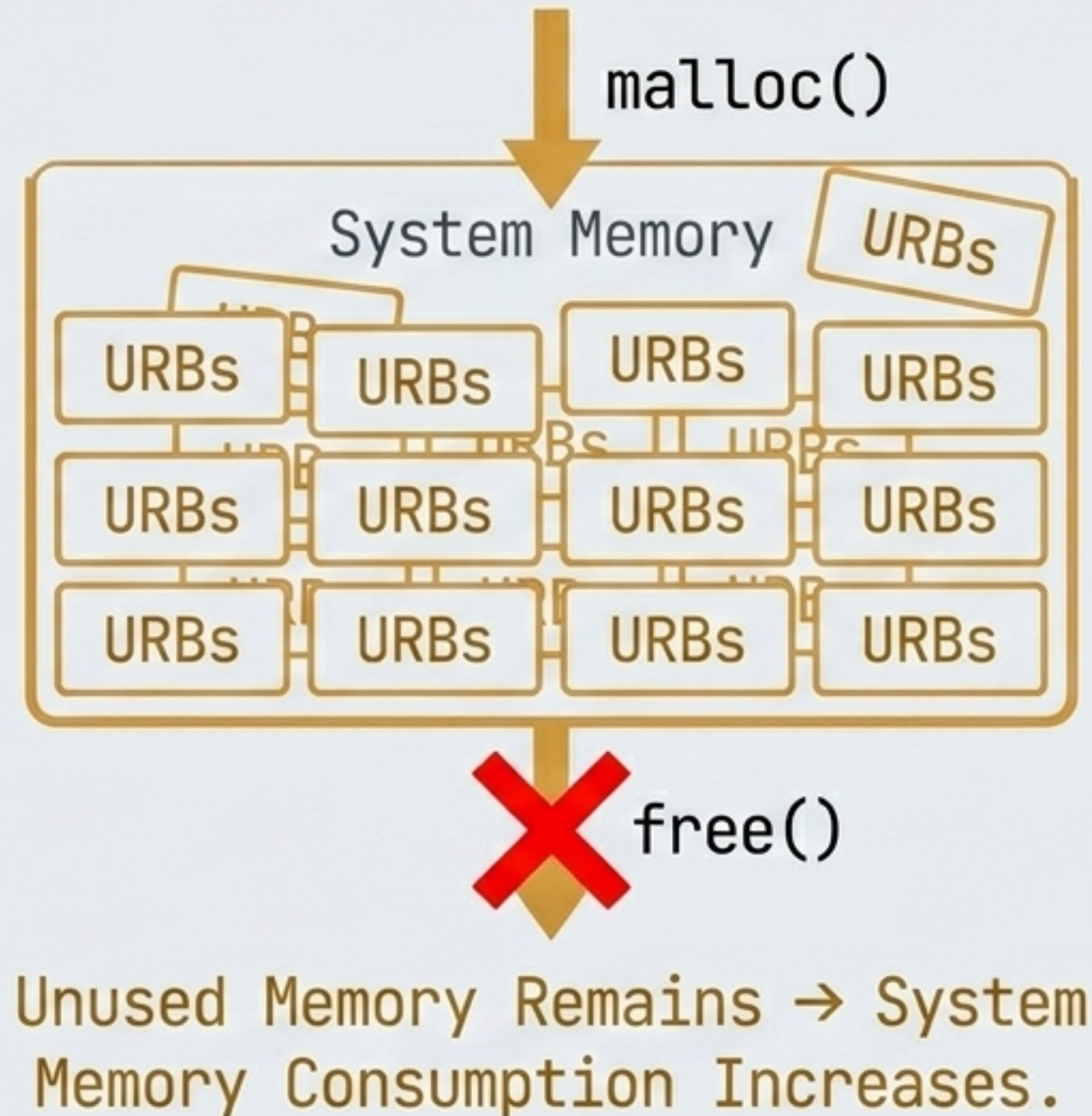


# The Anatomy of a Memory Leak

## HEALTHY STATE



## LEAK STATE





# The Root Cause: Partial Allocation Mishandling

Function: `es58x_alloc_rx_urbs()`

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## ✓ Expected Behavior

[ Partial URB Allocation ] → [ Continue Operation with successfully allocated URBs ]

---

## ⚠ Actual Behavior

[ Partial URB Allocation ] → [ Incorrectly Treat as Fatal Error ] → [ Exit Function Early ]



# Comparative Execution Flow

## Normal Flow

Allocate 10 URBs



Use URBs



Reach Cleanup Label



Free URBs (Success)

## Vulnerable Flow

Allocate 10 URBs



Partial Success (e.g., 8 allocated)



Driver Returns Error



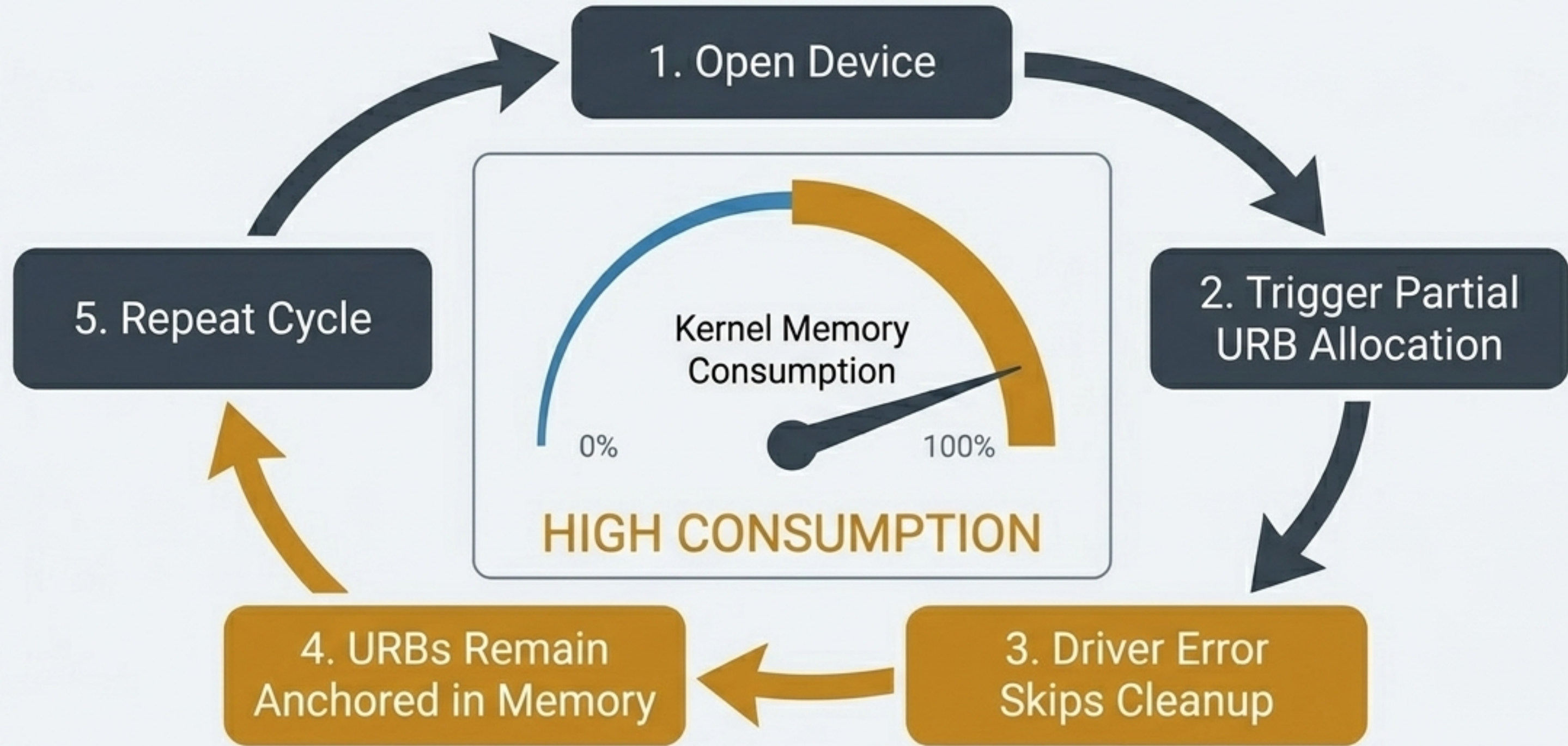
Cleanup Label Skipped



Memory Stranded (Leak 8 URBs)



# The Exploitation Engine: Compounding Impact



Repeated triggering turns a localized bug into systemic resource exhaustion.

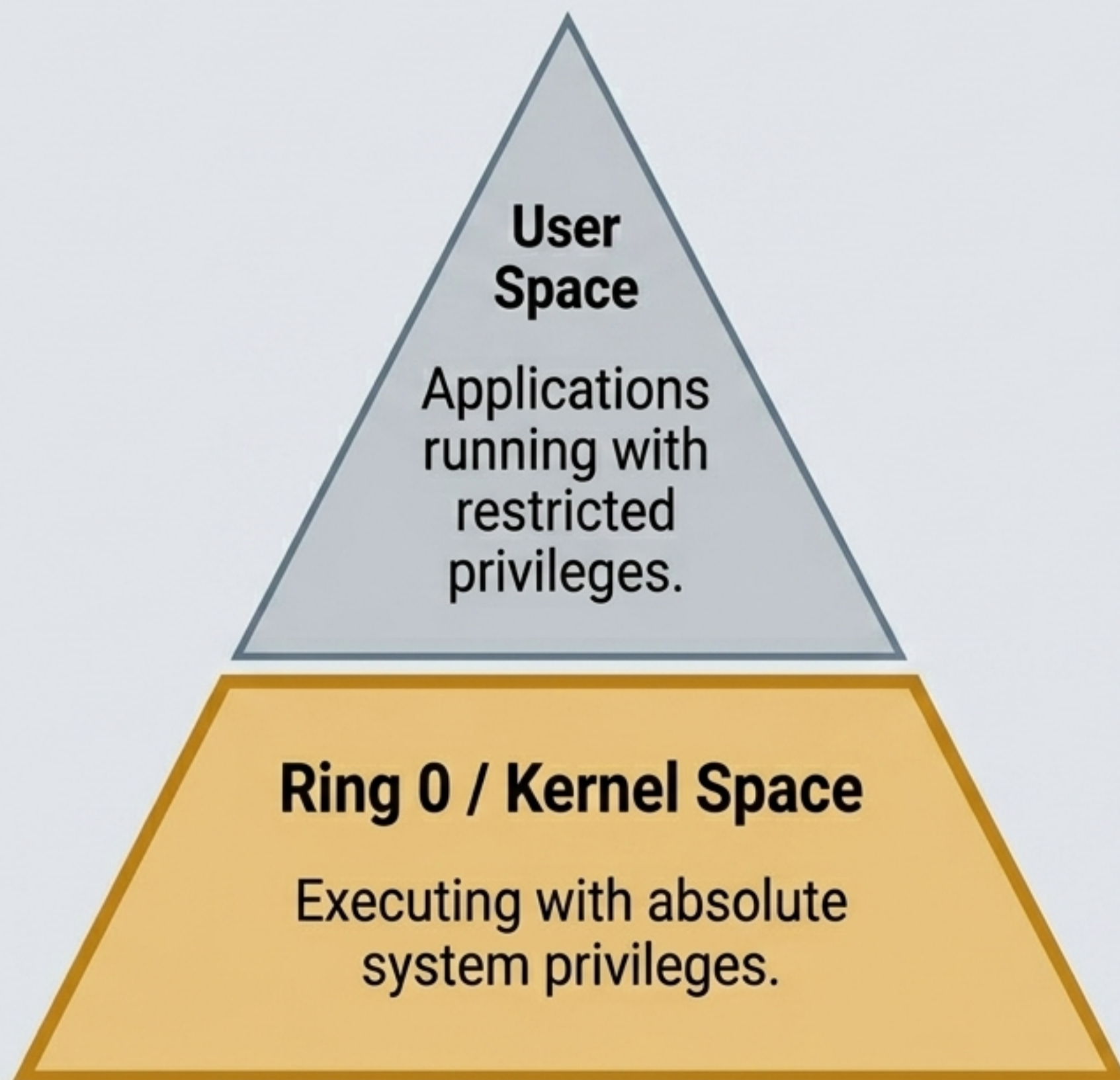


# Threat Profile & MITRE Matrix

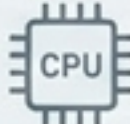
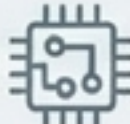
| Attack Prerequisites                                                                                                                          | CIA Impact Matrix                                                                                                                             | MITRE ATT&CK Mapping                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Required:</b> <ul style="list-style-type: none"><li>✓ Local Access</li><li>✓ Vulnerable Kernel</li><li>✓ ETAS ES58X Driver Usage</li></ul> | <b>Confidentiality:</b> None<br>(No Data Exposure)                                                                                            |  T1499 (Endpoint DoS)                   |
|                                                                                                                                               | <b>Integrity:</b> None<br>(No Modification)                                                                                                   |  T1496 (Resource Hijacking)           |
|                                                                                                                                               | <b>Availability:</b> Compromised<br>(Memory Exhaustion)  |  T1068 (Potential Kernel-Level Abuse) |



# Systemic Consequences: Why Kernel Bugs Hit Harder



Because kernel drivers execute with the highest system privileges, a memory leak here does not just crash an isolated application. It starves the entire operating system, resulting in:

-  Performance degradation
-  CAN device failures
-  Hardware instability
-  Ultimate kernel reliability issues (Panics/Freezes)



# Indicators of Compromise & Telemetry

```
> slabtop
```

```
> vmstat
```

```
> free -m
```

```
> cat /proc/meminfo
```

## Anomalies to Watch For



Increasing, unexplained Kernel Allocations



Persistent, uncleared USB Resources



Growing USB Driver Error logs



Unexpected CAN Device Failures



# Remediation Matrix & Security Controls

## Fixed Versions - Secure Kernels

5.15.199+

6.1.162+

6.6.122+

6.12.67+

6.18.7+

6.19+

## Compensating Controls



Apply relevant kernel updates



Monitor kernel memory telemetry via system tools



Disable unused CAN/USB drivers if not actively required by the environment



# Executive Synthesis

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CVE-2026-23037 turns a routine error-handling edge case (partial URB allocation) into a cumulative system threat. Because the ETAS ES58X driver skips its cleanup routine on partial failures, repeated triggering anchors memory in the kernel, inevitably leading to resource exhaustion. Applying updates to patched kernel versions fully resolves the underlying logic flaw.